

Project Proposal: Network-Augmented Latent Factor Portfolio — A Weekly-Rebalanced Long/Short Crypto Factor Strategy

Module: Quantitative Trading Strategy

Date: May 2026

1. Project Title

Network-Augmented Latent Factor Portfolio (NALFP): A Systematic Long/Short Cryptocurrency Factor Strategy with Network Topology and Adaptive Regime Blending

2. Executive Summary

This project develops and critically evaluates a weekly-rebalanced, long/short cryptocurrency factor strategy that integrates three pillars: (1) cross-sectional factor portfolios grounded in published asset-pricing literature, (2) time-varying network topology derived from minimum spanning trees and community detection on return correlation matrices, and (3) adaptive signal blending weighted by out-of-sample Information Coefficient (IC). The strategy is backtested over 48 weeks (24 in-sample, 24 out-of-sample) on a universe of approximately 80 crypto assets, with robust performance reporting, ablation analysis, and an honest assessment of a negative out-of-sample Sharpe ratio. The project openly documents where the strategy fails and why, in line with the principle that rigorous critical evaluation is more valuable than inflated backtested returns.

3. Investment Universe

The strategy operates on a dynamic universe of approximately 80–113 crypto assets, selected weekly as follows:

- **Eligibility:** Top assets by lagged market capitalisation, excluding stablecoins, wrapped tokens, and bridged assets (flagged via CoinGecko meta-data).
- **Data sources:** CoinGecko (daily OHLCV), Artemis on-chain activity (DAU, fees, revenue), and DeFiLlama (TVL, stablecoin flows).
- **Minimum history:** Assets must have 8 overlapping weeks with sufficient price history for correlation matrix construction.
- **Median portfolio size:** 16 long + 16 short positions per week (top and bottom quintiles).

This universe construction avoids survivorship bias by using lagged market cap for inclusion and excluding synthetic/stablecoin assets that would distort factor signals.

4. Factor Identification and Economic Justification

4.1 Core Factor Zoo

Nine economically-motivated long/short factor portfolios are constructed weekly. Each factor has a clear theoretical foundation drawn from published research in both traditional equities and crypto asset pricing:

Factor	Sort Variable	Direction	Economic Rationale	Key Reference
RC	Value-weighted market return	Market factor	Broad risk premium; analogous to equity market factor	Hartmann 2025
SMBC	log_mcap	Small-long, Big-short	Small-cap premium: compensation for higher information asymmetry, lower liquidity, and greater fundamental risk in crypto	Fama & French 1993; Hartmann 2025
MomC	mom_4w	Winners-long, Losers-short	Momentum: investors underreact to news; recent winners continue outperforming due to slow information diffusion and positive feedback trading	Jegadeesh & Titman 1993; Liu & Tsyvinski 2022
VolC	vol_4w	Low-vol long, High-vol short	Betting-against-beta: leverage-constrained investors overbid high-beta assets, depressing their risk-adjusted returns	Frazzini & Pedersen 2014

Factor	Sort Variable	Direction	Economic Rationale	Key Reference
TVLC	<code>tv1_to_mcap</code>	High-TVL long, Low-TVL short	Network value: high TVL-to-market-cap ratios signal genuine usage relative to speculative valuation	Hartmann 2025
FunC	<code>F_yield</code>	High-yield long, Low-yield short	Fundamental yield: crypto analogue of earnings yield (E/P); protocols generating real fees relative to market cap	RAAM v2 (this project)
SupC	<code>S_supply</code>	Low-emission long, High-emission short	Supply absorption: tokens with low net new supply face less structural selling pressure	RAAM v2 (this project)
NetMom	<code>within_cluster_mom</code>	Top-half long, Bottom-half short (within cluster)	Cluster-neutral momentum: isolates idiosyncratic momentum from broad-market trend by ranking within Louvain communities	Liu & Tsyvinski 2018
NetRel	<code>cross_cluster_rel</code>	Outperformers long, Underperformers short	Cross-cluster relative strength: captures narrative rotation between crypto thematic clusters	Liu & Tsyvinski 2018

4.2 Network Topology Pillar

Beyond static cross-sectional factors, the strategy incorporates time-varying network structure as a novel signal source:

- **Construction:** Each week, a 12-week rolling Spearman correlation matrix is computed across all eligible assets. This is converted to a Man-

tegrate distance matrix, from which a Minimum Spanning Tree (MST) is extracted. Louvain community detection (resolution = 1.0) then partitions the MST into 7–11 thematic clusters per week.

- **Signals:** Three per-asset signals (within-cluster momentum, cross-cluster relative strength, cluster label) and one market-level signal (Shannon entropy of cluster-size distribution as a fragmentation proxy).
- **Economic thesis:** When the correlation network is fragmented (many small clusters), narratives are localised and within-cluster relative performance is more informative. When the network converges, common latent factors dominate. The adaptive blend operationalises this thesis without discretionary parameters.

4.3 Latent Factor Pricing (Giglio–Xiu Three-Pass Procedure)

To address omitted-variable bias in observed factor premia, the strategy applies the Giglio & Xiu (2021) three-pass procedure in walk-forward mode:

1. **Pass 1:** Time-series regression of each asset’s weekly return on the 7 observed factors with sufficient coverage.
2. **Pass 2:** PCA on the residual matrix; Bai–Ng ICp2 criterion selects Khindden = 3 latent factors.
3. **Pass 3:** Cross-sectional Fama–MacBeth estimation of risk premia $\hat{\alpha}$ on the combined observed + latent factor set, with heteroskedasticity-robust (White) standard errors.

This produces per-asset expected returns ($E_{gx,it}$) that are debiased for latent factor exposure — a significant methodological improvement over raw factor sort returns.

4.4 Adaptive Regime Blend

The final signal combines the network momentum stream and the GX expected-return stream using out-of-sample IC-proportional weighting:

$$E_{final,it} = w_{net,t} \cdot E_{net,it} + (1 - w_{net,t}) \cdot E_{gx,it}$$

where $w_{net,t}$ is set by the rolling 8-week mean Spearman IC of each stream against next-week returns. If both ICs are non-positive, the weight defaults to 0.5. This is a mechanical, parameter-free blend — no optimisation or discretionary choices.

5. Portfolio Construction Methodology

Step	Rule	Rationale
Selection	Top 20% long, bottom 20% short (quintile cut on E_{final})	Extracts signal from cross-sectional dispersion while limiting portfolio size
Position sizing	Inverse 4-week realised volatility within each leg, normalised to long = +1, short = -1 (dollar-neutral)	Low-vol names receive larger weights; dollar-neutral construction removes directional market exposure
Single-asset cap	Maximum 5% of gross portfolio per name; excess redistributed pro-rata (iterative, up to 10 rounds)	Prevents concentration in any single asset
Cluster diversification cap	Long-leg exposure to any single Louvain cluster 40%	Limits thematic overconcentration; tested via ablation
Turnover budget	Weekly one-sided turnover 30%; smallest	Δw
Vol target	Gross leverage scaled to target 15% annualised realised volatility, capped at $3\times$ gross	Risk-management overlay; constrains portfolio to pre-specified risk budget

The strategy rebalances weekly on Monday at market close.

6. Backtest Design and Implementation

6.1 Walk-Forward Framework

Parameter	Value	Justification
Total sample	48–49 weeks (after 12-week network burn-in)	52-week full panel minus burn-in
Training window	24 weeks	Expanding window for factor premia estimation

Parameter	Value	Justification
OOS evaluation	24 weeks	Holdout for genuine out-of-sample performance
Re-estimation frequency	Every 4 weeks	Walk-forward; expanding window, not rolling
Cost model	10 bps one-sided per unit of weekly turnover	Conservative for median-name liquidity; $= 0.5 \times \Sigma$
Vol annualisation	$\sqrt{52}$ (weekly returns)	Standard for weekly return compounding
Bootstrap	Stationary block bootstrap, block length 4 weeks, 2,000 draws	For Sharpe ratio confidence intervals

6.2 No-Lookahead Guarantees

- All characteristics are lagged one week prior to entering predictive models.
- Portfolio weights at week t use only information available at $t - 1$.
- Factor premia are estimated on expanding training windows only.
- The 12-week correlation window for network construction uses only historical returns.

6.3 Benchmarks

Three benchmark strategies are evaluated alongside NALFP:

1. **Equal-Weight Momentum Long** — naive long-only top-quintile momentum strategy (no short leg, no vol targeting).
2. **RAAM v2 Composite** — IC-weighted multi-signal composite from the upstream factor signal stage.
3. **plus_lat** — Fama–MacBeth predictions from the stage-06 econometrics model (characteristics + PCA latent controls).

6.4 Summary Statistics Reported

For all strategies and factors:

- Annualised return, annualised volatility, Sharpe ratio
- Maximum drawdown
- Average weekly turnover
- Hit rate (% of positive-return weeks)
- Newey–West t-statistics on factor premia

- Bootstrap 95% confidence intervals for Sharpe ratios
- Cross-sectional Information Coefficient (IC) and IC Information Ratio (IR)

6.5 Ablation Analysis

An ablation variant removes the cluster diversification cap (`apply_cluster_cap=False`) to test whether the network pillar’s operational constraint materially affects performance or merely serves as risk management.

7. Alignment with Module Assessment Criteria

Criterion	How This Project Addresses It
(i) Fundamental economic justification	Each of the nine factors is grounded in published asset-pricing literature. The network pillar is motivated by the thesis that crypto market structure is organised into thematic communities rather than a single latent factor. The adaptive blend is theoretically motivated by the interaction between network fragmentation and factor dominance. Factor premia are estimated via the Giglio–Xiu three-pass procedure, which corrects for omitted-variable bias — not merely pattern-fitting.
(ii) Implementation of actual strategy	The strategy is fully implemented in a reproducible Python pipeline (six sequential scripts). Portfolio construction includes position sizing, concentration limits, cluster diversification caps, turnover budgets, and volatility targeting. All code is version-controlled and artefacts are archived as parquet/CSV/JSON manifests.
(iii) Quality and realism of backtest	Walk-forward estimation with expanding training windows; no lookahead bias; 10 bps transaction costs; bootstrap confidence intervals for Sharpe ratios; ablation analysis testing the marginal impact of the cluster cap; three benchmark strategies for comparison; honest reporting of negative OOS performance.
(iv) Discussion of performance metrics	Comprehensive reporting of Sharpe ratios, maximum drawdown, turnover, hit rate, Newey–West t-statistics, IC/IR, and bootstrap CIs. An entire section is devoted to what failed and why, with structural limitations explicitly catalogued. The negative OOS Sharpe is not hidden — it is analysed in depth as the most important finding of the project.

8. References

- Bai, J. & Ng, S. (2002). “Determining the Number of Factors in Approximate Factor Models.” *Econometrica*, 70(1), 191–221.
- Carhart, M. (1997). “On Persistence in Mutual Fund Performance.” *Journal of Finance*, 52(1), 57–82.
- Fama, E. & French, K. (1993). “Common Risk Factors in the Returns on Stocks and Bonds.” *Journal of Financial Economics*, 33(1), 3–56.
- Frazzini, A. & Pedersen, L. (2014). “Betting Against Beta.” *Journal of Financial Economics*, 111(1), 1–25.
- Giglio, S. & Xiu, D. (2021). “Portfolio Selection with Asset Pricing Factors.” *Review of Financial Studies*, 34(8), 3816–3847.
- Hartmann, B. (2025). “Crypto Factors.” Working paper.
- Jegadeesh, N. & Titman, S. (1993). “Returns to Buying Winners and Selling Losers.” *Journal of Finance*, 48(1), 65–91.
- Kelly, B., Pruitt, S. & Su, Y. (2019). “Characteristics Are Covariances.” *Journal of Finance*, 74(4), 1809–1854.
- Liu, Y. & Tsyvinski, A. (2022). “Risks and Returns of Cryptocurrency.” *Review of Financial Studies*, 34(6), 2689–2727.
- Mantegna, R. (1999). “Hierarchical Structure in Financial Markets.” *European Physical Journal B*, 11(1), 193–197.